MUTUAL FUND ANALYTICS & INVESTOR INSIGHTS PLATFORM

Final Project Report

BLUESTOCK™.in

Prepared By

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Technologies Used

Python, SQL, SQLite, Power BI, Pandas, NumPy, Matplotlib, GitHub

Executive Summary

The Mutual Fund Analytics & Investor Insights Platform is an end-to-end data analytics project designed to analyze mutual fund performance, investor behavior, SIP trends, and portfolio risk. The project integrates multiple financial datasets and applies data engineering, exploratory analysis, financial performance evaluation, advanced analytics, and business intelligence techniques.

The project was developed using Python, SQL, SQLite, and Power BI. It includes data ingestion, cleaning, transformation, dashboard development, and advanced analytical modules such as VaR, CVaR, Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation Engine, and Portfolio Concentration Analysis.

The final output includes an interactive Power BI dashboard, advanced analytical reports, automated processing scripts, and actionable business insights.

Business Problem

The mutual fund industry generates large volumes of data related to fund performance, investor transactions, SIP investments, portfolio holdings, and market benchmarks. Investors often face challenges in identifying suitable funds, understanding risk, evaluating performance, and monitoring investment behavior.

Financial institutions require analytical solutions that can:

- Monitor fund performance
- Understand investor behavior
- Evaluate portfolio risk
- Track SIP growth
- Generate investment insights
- Support informed decision-making

This project addresses these requirements through a comprehensive analytics platform.

Project Objectives

The key objectives of the project were:

1. Build a structured ETL pipeline for mutual fund data.
2. Clean and standardize multiple datasets.
3. Perform exploratory data analysis.
4. Calculate advanced performance metrics.
5. Develop a multi-page Power BI dashboard.
6. Analyze investor behavior and transaction trends.
7. Build a simple fund recommendation engine.
8. Generate actionable business insights.

Data Sources

The project uses ten datasets:

Fund Master

Contains scheme information, fund house details, benchmark, category, expense ratio, and risk category.

NAV History

Historical Net Asset Value (NAV) records used for return calculations.

AUM by Fund House

Assets Under Management information for major fund houses.

Monthly SIP Inflows

Industry-level SIP investment data.

Category Inflows

Category-wise investment inflows across different mutual fund categories.

Industry Folio Count

Investor participation measured through folio counts.

Scheme Performance

Fund performance metrics and return information.

Investor Transactions

Investor-level transaction records including SIPs, lumpsum investments, and redemptions.

Portfolio Holdings

Sector-wise portfolio allocation data.

Benchmark Indices

Historical NIFTY50 and NIFTY100 benchmark values.

Project Architecture

The project follows a structured analytics workflow:

Raw Data Sources

↓

Data Ingestion

↓

Data Cleaning & Validation

↓

Feature Engineering

↓

SQLite Database

↓

Exploratory Data Analysis

↓

Performance Analytics

↓

Advanced Analytics

↓

Power BI Dashboard

↓

Business Insights

ETL Design

Extract

Data was collected from multiple CSV datasets containing mutual fund and investor information.

Transform

Several transformation activities were performed:

- Missing value handling
- Duplicate removal
- Date standardization
- Numeric conversion
- Data validation
- Feature engineering
- Data quality checks

Load

Cleaned datasets were loaded into SQLite for further analysis and reporting.

Data Cleaning and Validation

Several quality issues were identified and resolved:

Missing Values

Missing values were detected and handled appropriately using imputation and validation techniques.

Duplicate Records

Duplicate entries were removed to ensure consistency.

Data Type Corrections

Date and numeric columns were converted into appropriate formats.

Validation Checks

Business validation rules were applied to identify anomalies and inconsistencies.

These steps improved overall data reliability and analysis quality.

Exploratory Data Analysis

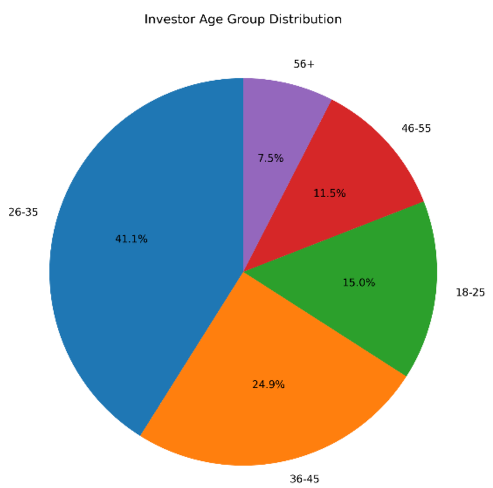
Exploratory analysis was conducted to understand mutual fund industry trends and investor behavior.

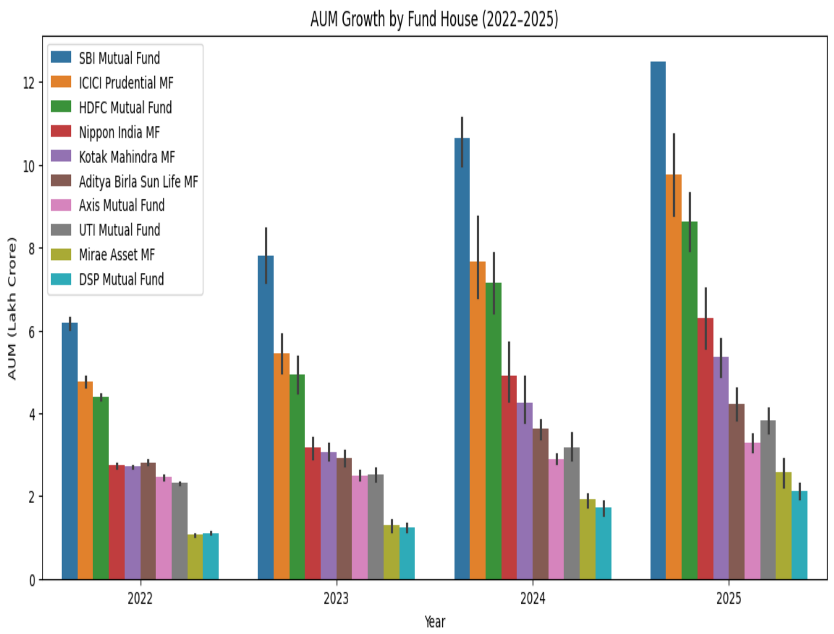
The analysis focused on:

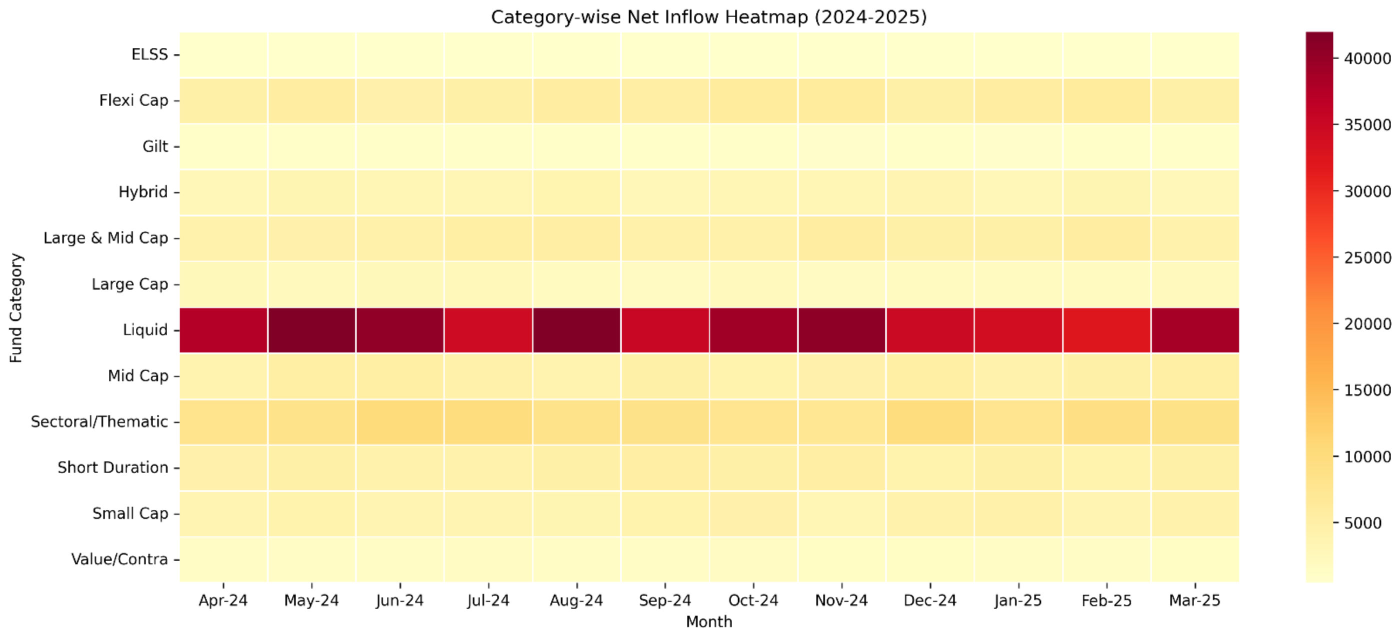
- NAV growth patterns
- Fund house performance
- SIP growth trends
- Investor demographics
- Geographic distribution
- Category inflows
- Portfolio composition

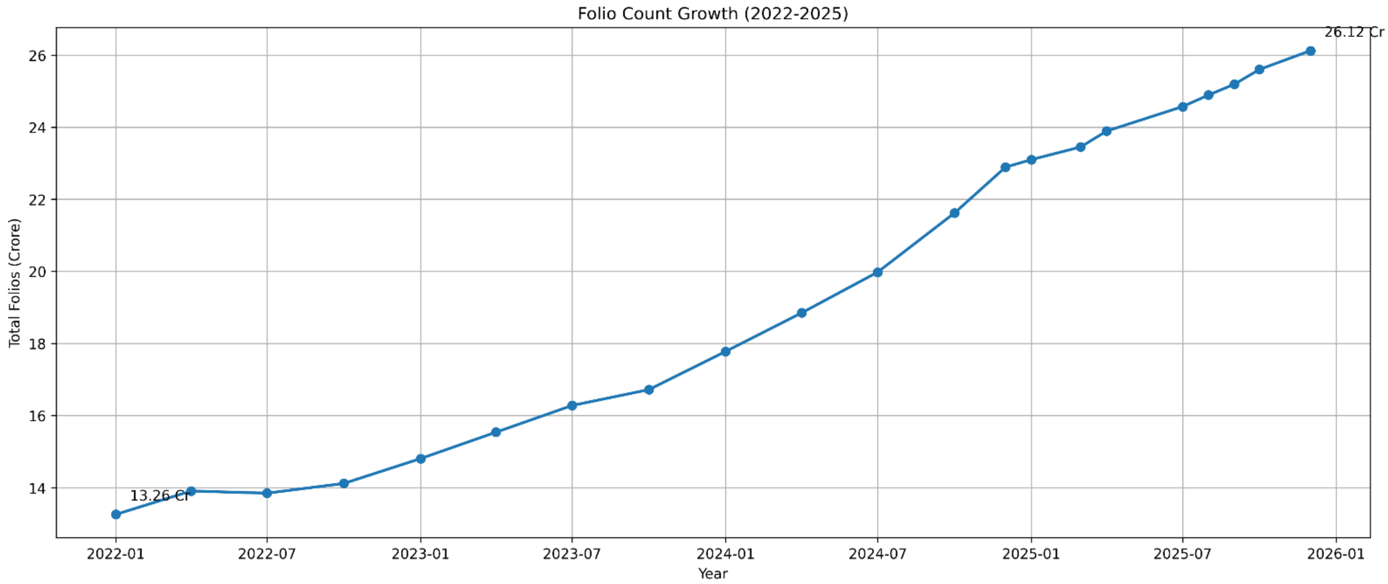
Key EDA Findings

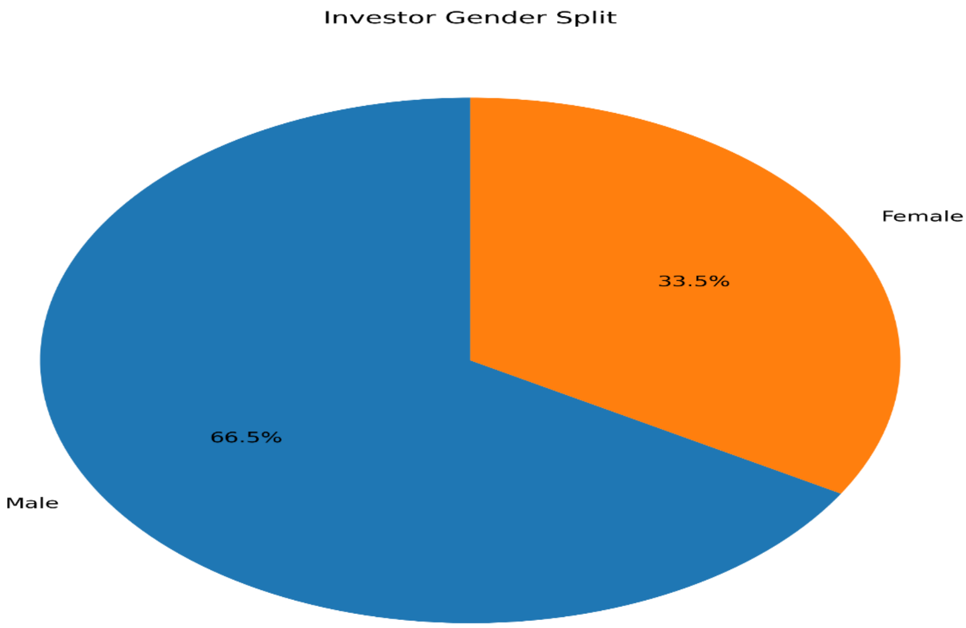
- Industry AUM increased significantly over time.
- SIP inflows reached record highs.
- Equity-oriented funds attracted higher inflows.
- Investor participation increased steadily.
- Certain states contributed disproportionately higher investment volumes.

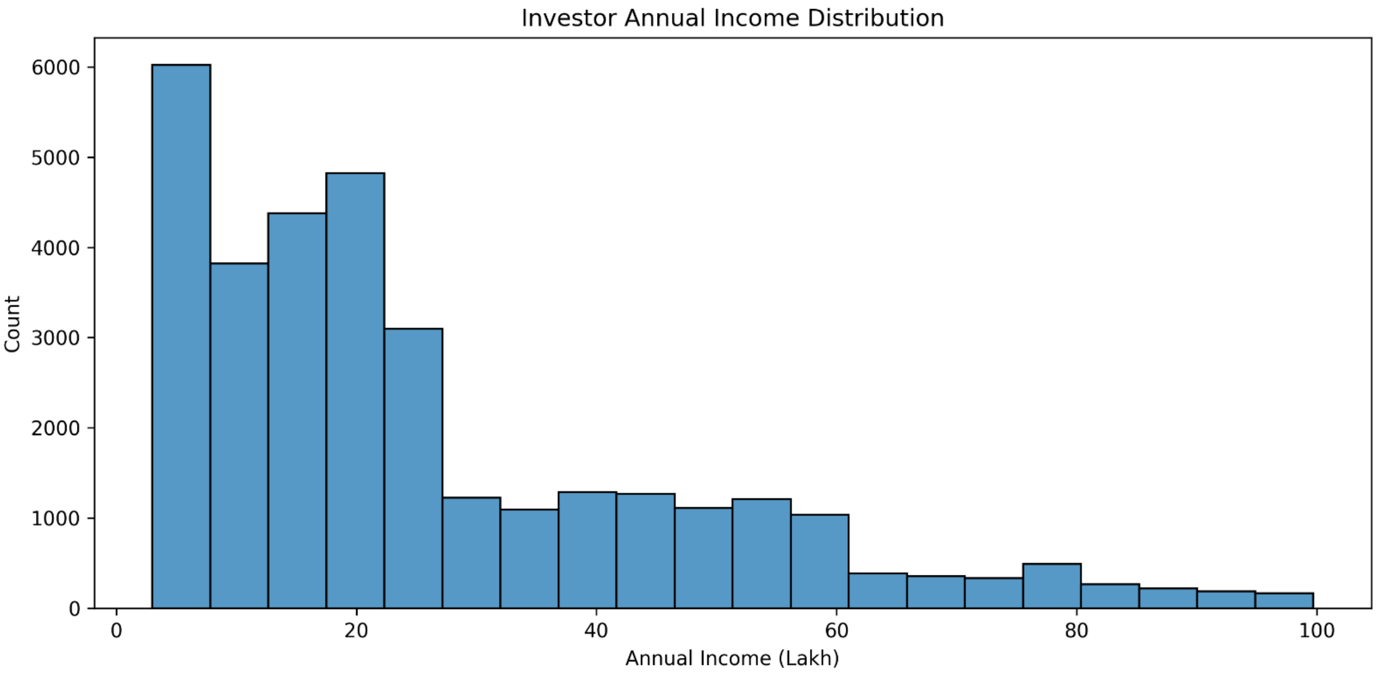


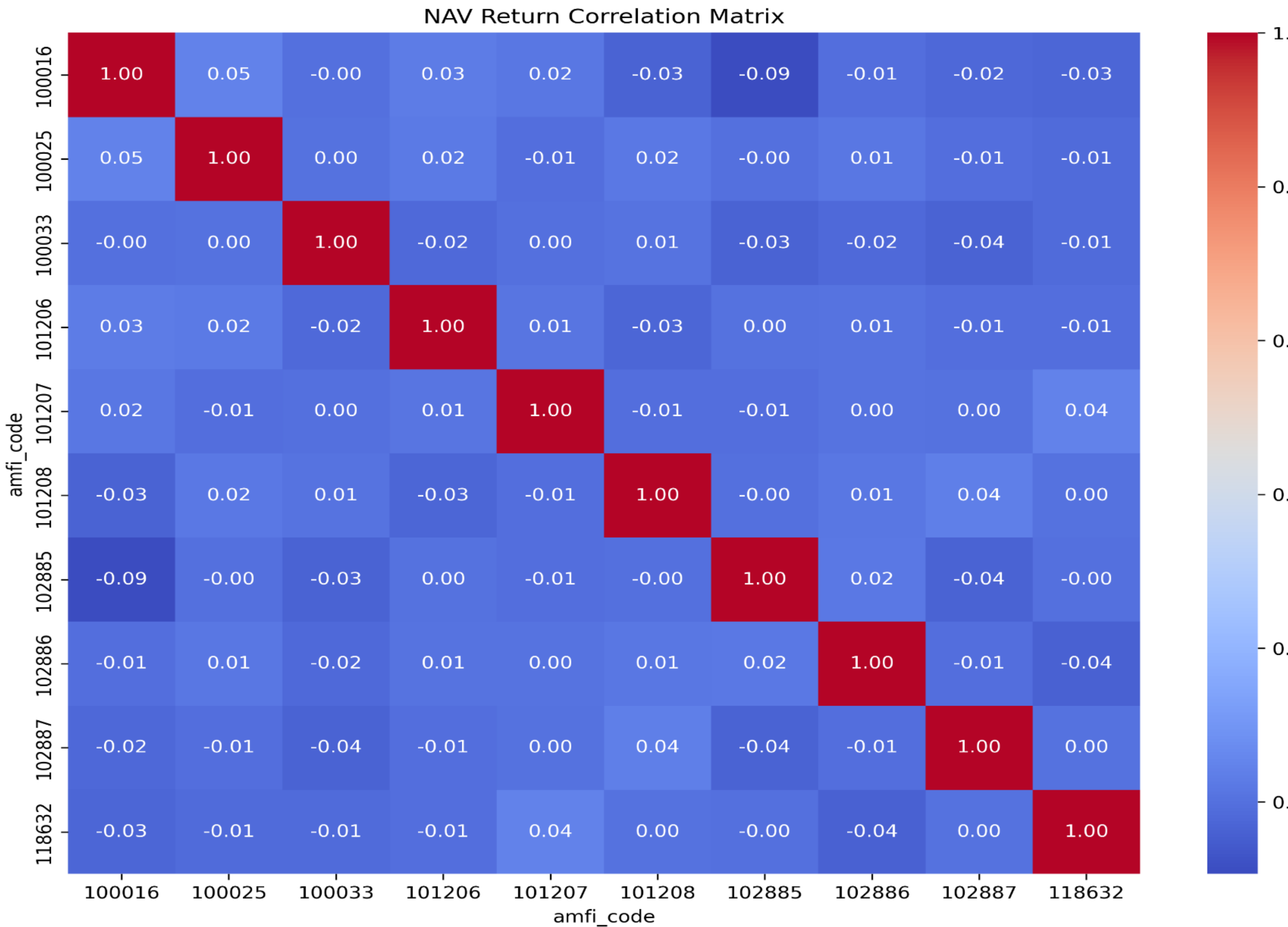


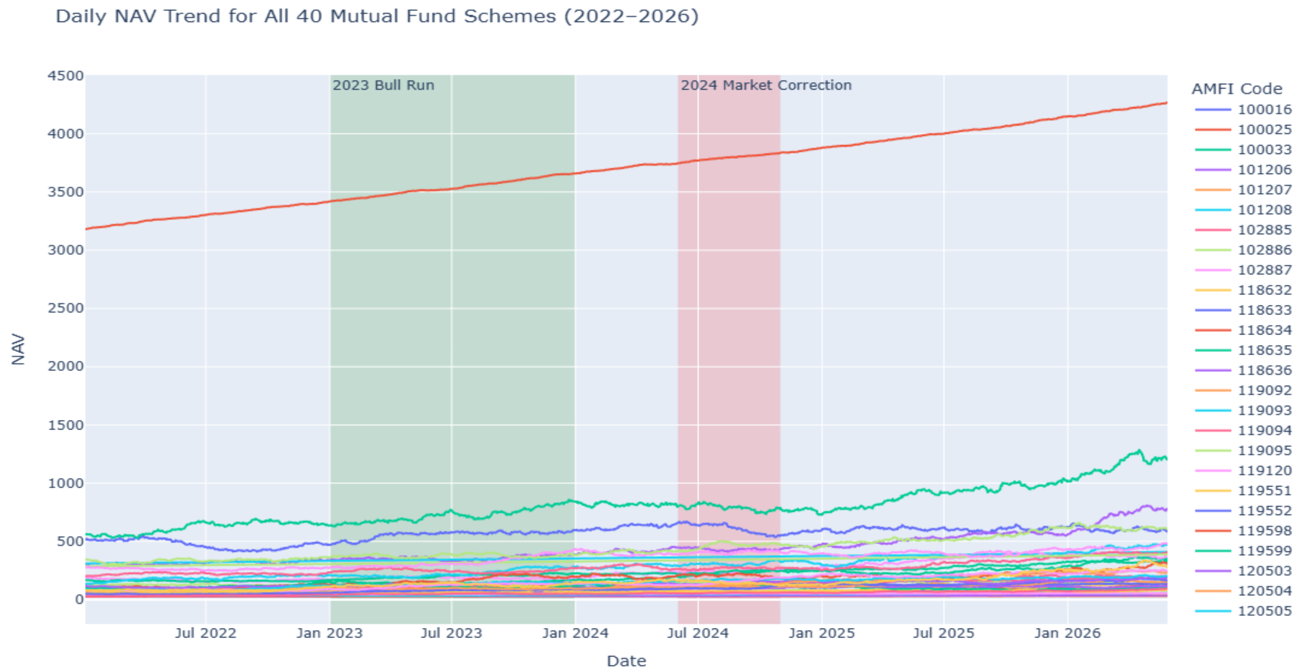

Category-wise Net Inflow Heatmap (2024-2025)
ELSS
Flexi Cap
Gilt
Hybrid
Large & Mid Cap
Large Cap
Liquid
Mid Cap
Sectoral/Thematic
Short Duration
Small Cap
Value/Contra
Fund Category
Apr-24 May-24 Jun-24 Jul-24 Aug-24 Sep-24 Oct-24 Nov-24 Dec-24 Jan-25 Feb-25 Mar-25
Month
40000
35000
30000
25000
20000
15000
10000
5000

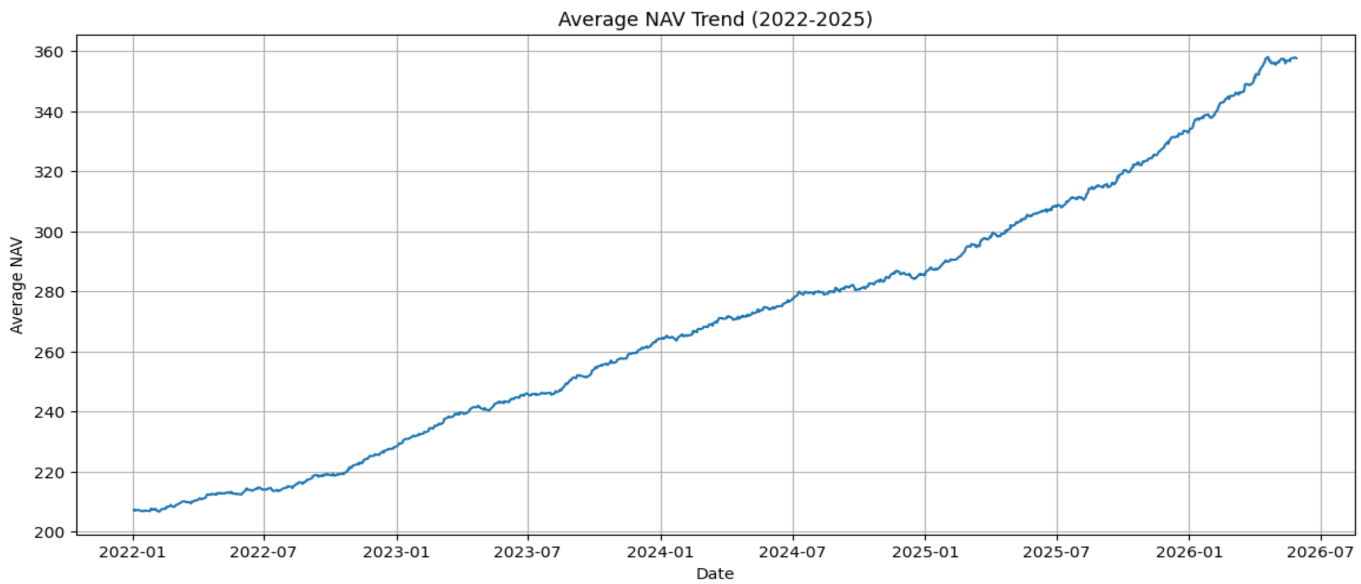

Folio Count Growth (2022-2025)
26.12 Cr
13.26 Cr
Total Folios (Crore)
14
16
18
20
22
24
26
2022-01 2022-07 2023-01 2023-07 2024-01 2024-07 2025-01 2025-07 2026-01
Year

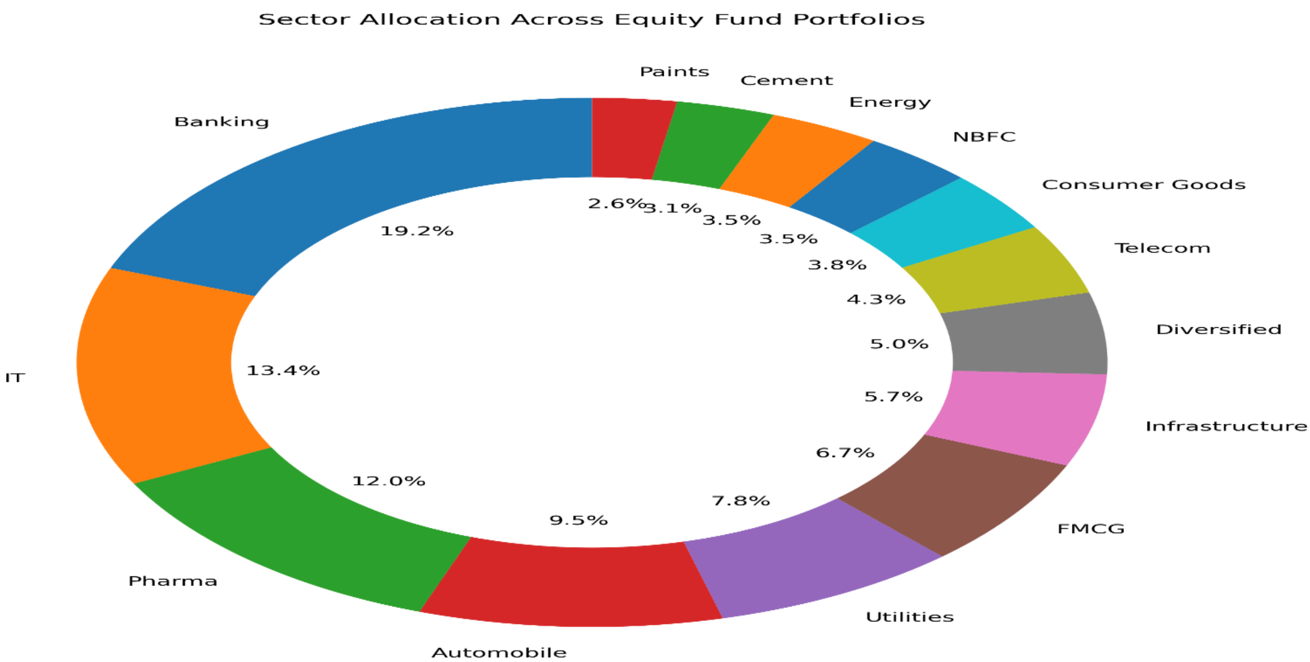

Investor Gender Split
Female
33.5%
66.5%
Male

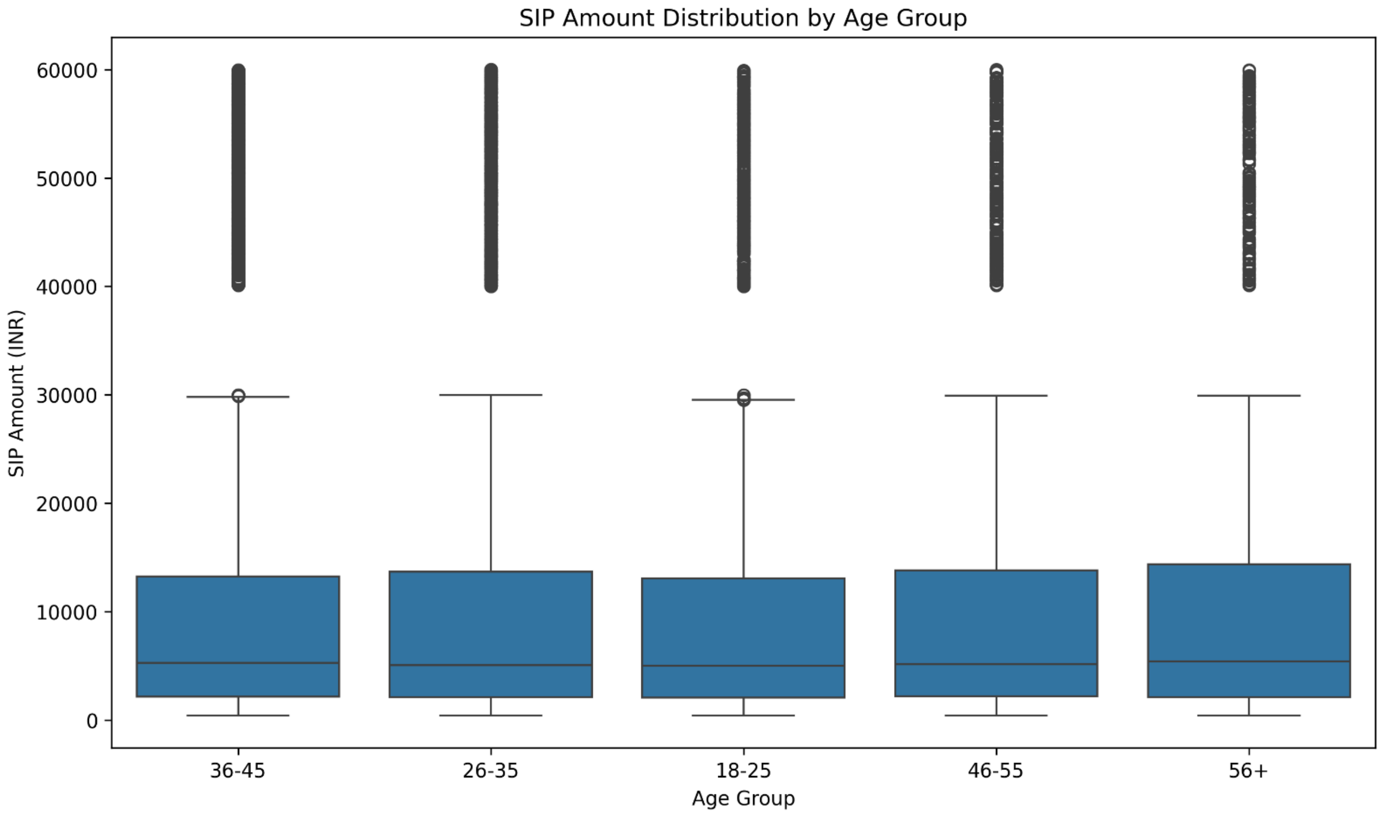

Investor Annual Income Distribution
Count
Annual Income (Lakh)
0
1000
2000
3000
4000
5000
6000
0
20
40
60
80
100


NAV Return Correlation Matrix
amfi_code
amfi_code
100016
100025
100033
101206
101207
101208
102885
102886
102887
118632

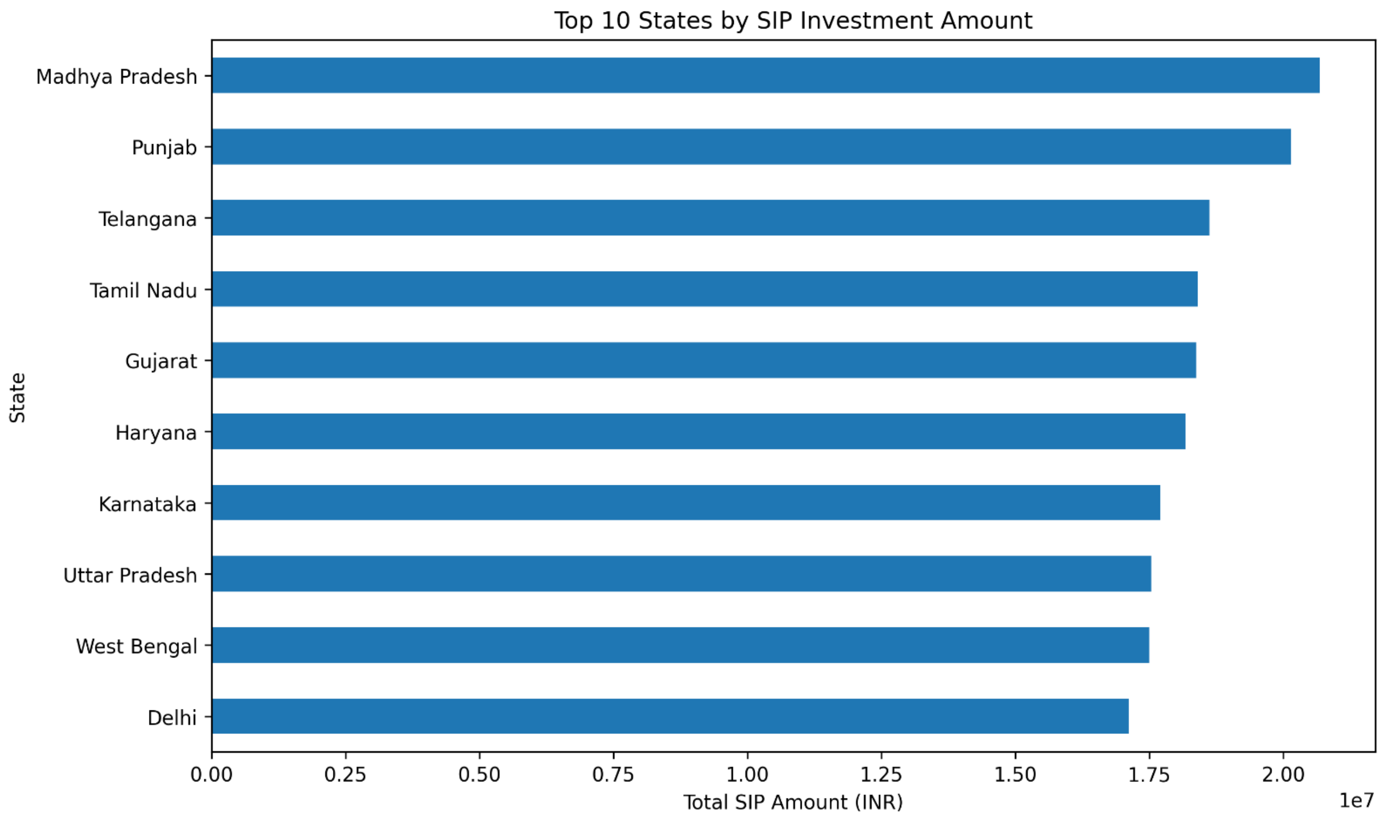

Daily NAV Trend for All 40 Mutual Fund Schemes (2022–2026)
2023 Bull Run
2024 Market Correction
NAV
Date
AMFI Code
100016
100025
100033
101206
101207
101208
102885
102886
102887
118632
118633
118634
118635
118636
119092
119093
119094
119095
119120
119551
119552
119598
119599
120503
120504
120505


Average NAV Trend (2022-2025)
Average NAV
Date

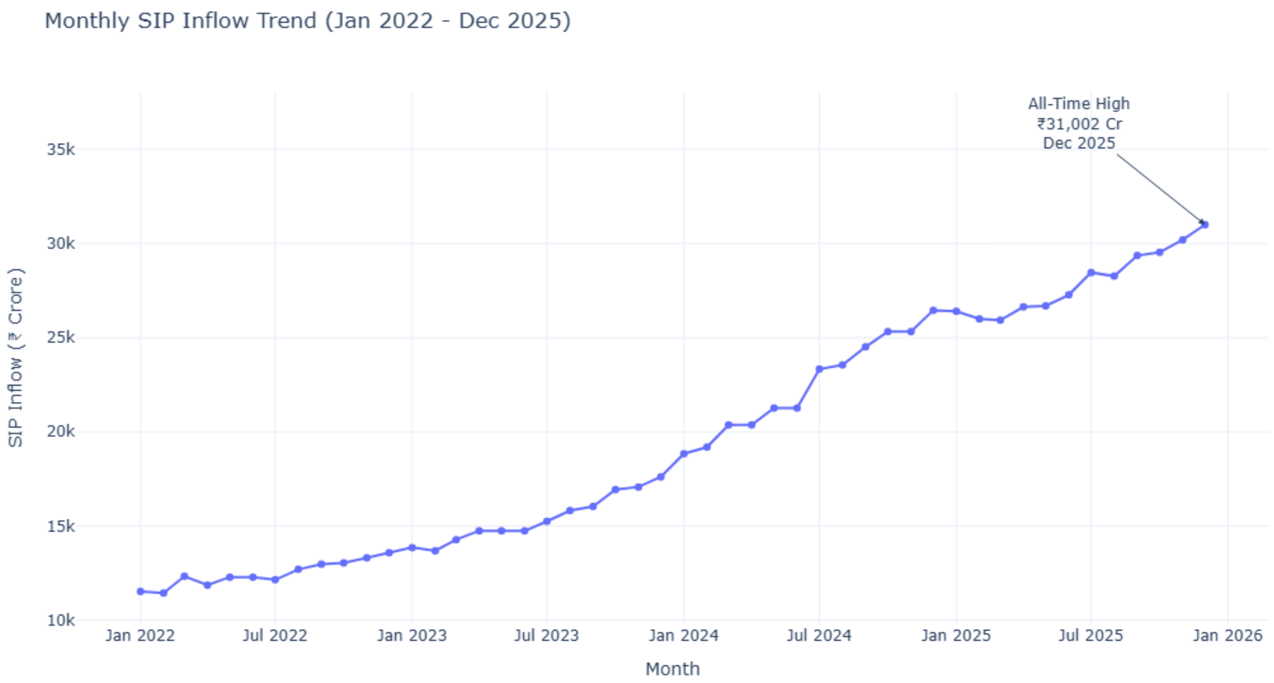

Sector Allocation Across Equity Fund Portfolios
Banking 19.2%
IT 13.4%
Pharma 12.0%
Automobile 9.5%
Utilities 7.8%
FMCG 6.7%
Infrastructure 5.7%
Diversified 5.0%
Telecom 4.3%
Consumer Goods 3.8%
NBFC 3.5%
Energy 3.5%
Cement 3.1%
Paints 2.6%



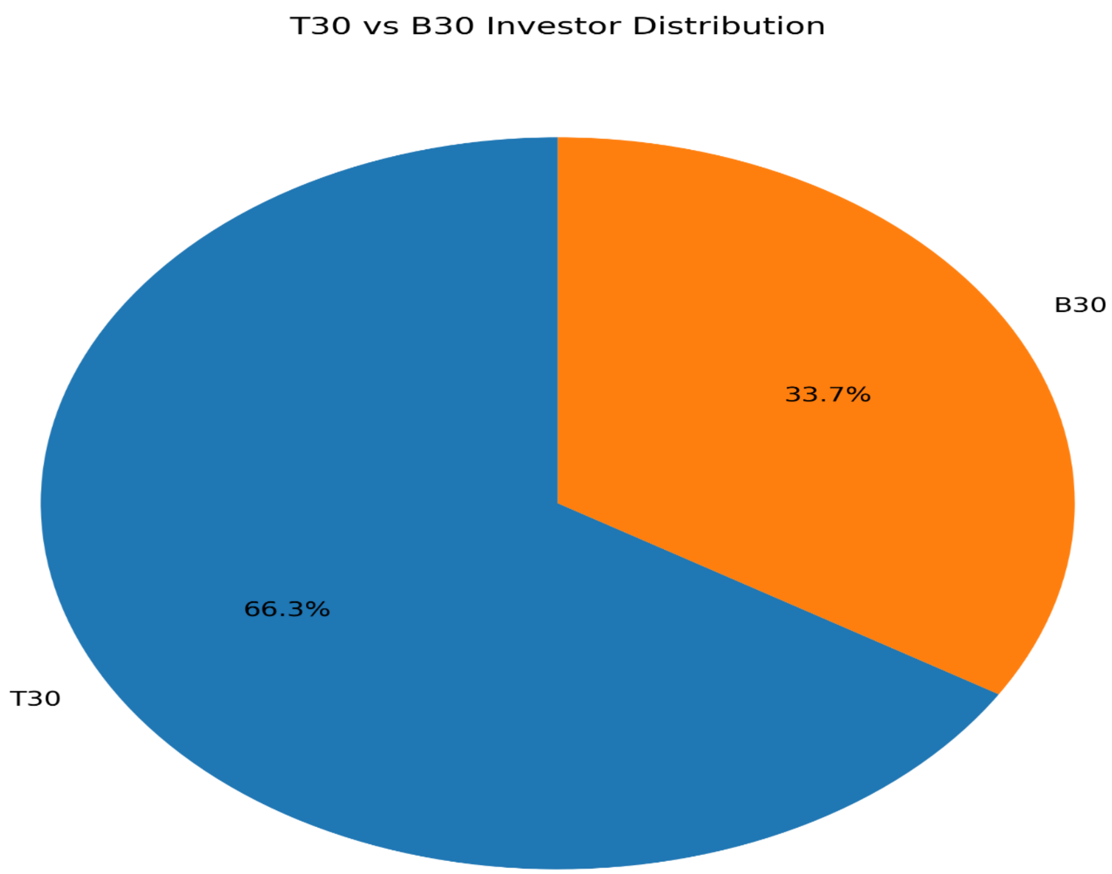
SIP Amount Distribution by Age Group
SIP Amount (INR)
60000
50000
40000
30000
20000
10000
0
36-45
26-35
18-25
46-55
56+
Age Group



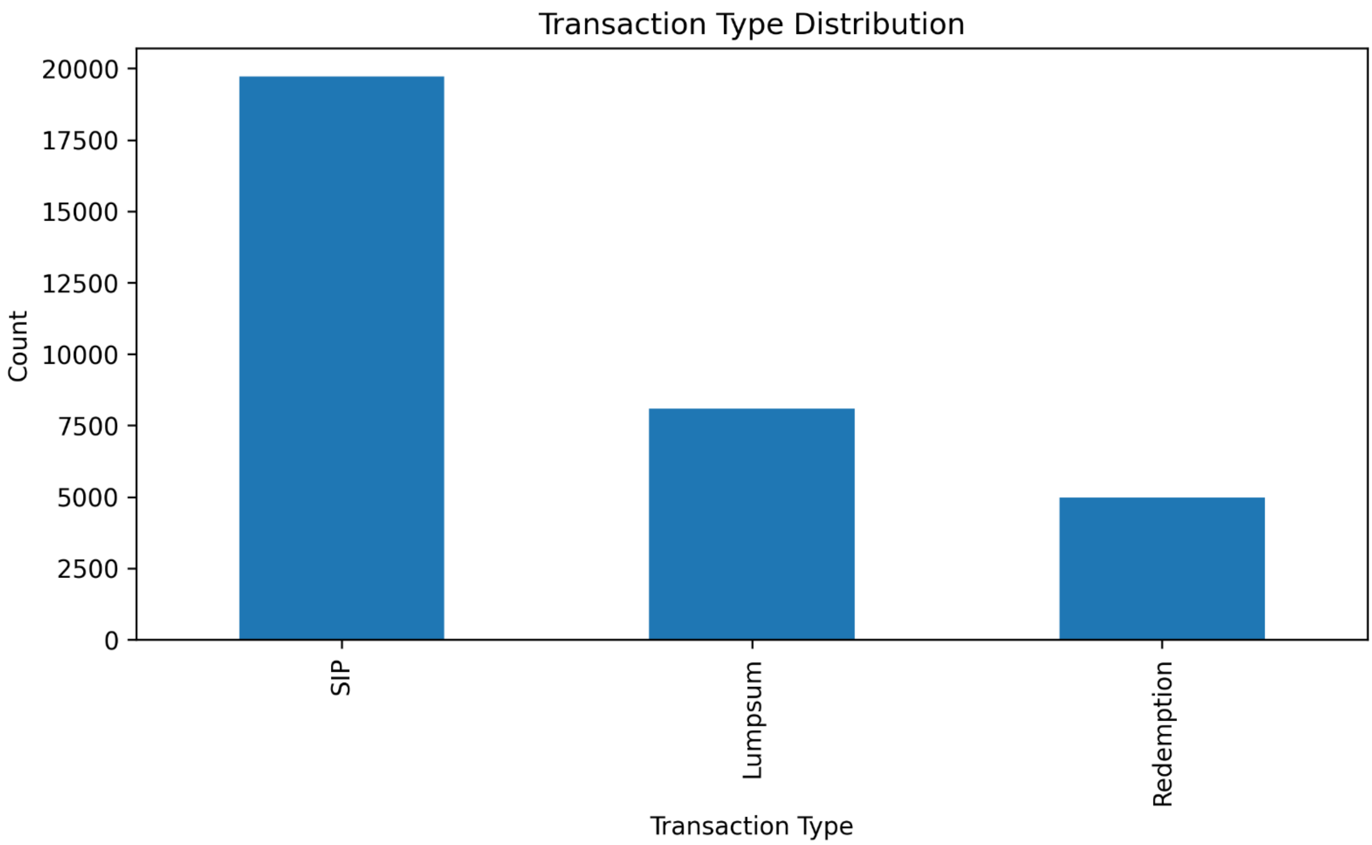
Top 10 States by SIP Investment Amount
State
Madhya Pradesh
Punjab
Telangana
Tamil Nadu
Gujarat
Haryana
Karnataka
Uttar Pradesh
West Bengal
Delhi
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
1e7
Total SIP Amount (INR)



Monthly SIP Inflow Trend (Jan 2022 - Dec 2025)
All-Time High
₹31,002 Cr
Dec 2025
SIP Inflow (₹ Crore)
35k
30k
25k
20k
15k
10k
Jan 2022
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Jan 2025
Jul 2025
Jan 2026
Month



T30 vs B30 Investor Distribution
B30
33.7%
66.3%
T30


Transaction Type Distribution
Count
20000
17500
15000
12500
10000
7500
5000
2500
0
SIP
Lumpsum
Redemption
Transaction Type

Performance Analytics

Performance analytics were developed to evaluate mutual fund performance using financial metrics.

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated for 1-year, 3-year, and 5-year periods.

The analysis identified top-performing funds with strong long-term growth.

Sharpe Ratio

Sharpe Ratio measured risk-adjusted performance.

Funds with higher Sharpe Ratios generated better returns relative to volatility.

Alpha and Beta

Alpha measured excess returns generated beyond benchmark expectations.

Beta measured sensitivity to market movements.

Sortino Ratio

Sortino Ratio evaluated downside-risk-adjusted performance.

Maximum Drawdown

Maximum Drawdown measured the largest decline from peak portfolio value.

Fund Scorecard

A composite score was developed using:

- CAGR
- Sharpe Ratio
- Alpha
- Expense Ratio
- Drawdown

This enabled ranking of all schemes.

Advanced Analytics

Historical VaR and CVaR

Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated for all 40 schemes.

Small-cap funds exhibited higher downside risk compared to large-cap funds.

Rolling Sharpe Ratio

Rolling 90-Day Sharpe Ratio was calculated to evaluate changing risk-adjusted performance over time.

Performance fluctuated significantly during different market conditions.

Investor Cohort Analysis

Investors were grouped based on their first transaction year.

The 2024 cohort represented the largest investor group and contributed the highest total investment value.

SIP Continuity Analysis

Investors with at least six SIP transactions were evaluated.

Accounts with an average transaction gap greater than 35 days were classified as At Risk.

The analysis highlighted opportunities for investor retention initiatives.

Fund Recommendation Engine

A rule-based recommendation system was developed using:

- Risk Category
- Sharpe Ratio

The system recommends the top three funds for each investor risk profile.

Portfolio Concentration Analysis

The Herfindahl-Hirschman Index (HHI) was used to measure sector concentration.

Funds with lower HHI values demonstrated stronger diversification.

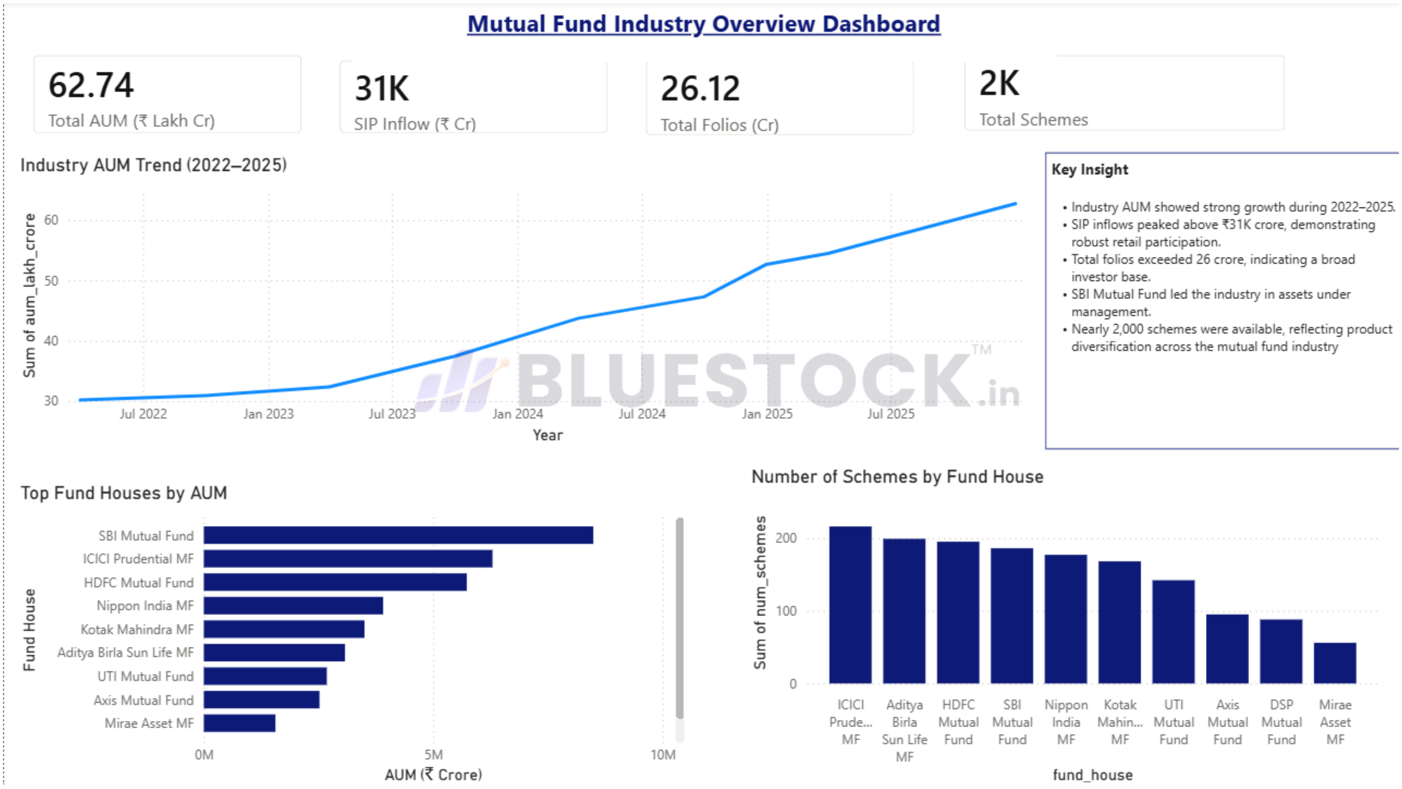
Power BI Dashboard

The final dashboard consists of four interactive pages.

Page 1 – Industry Overview

Features:

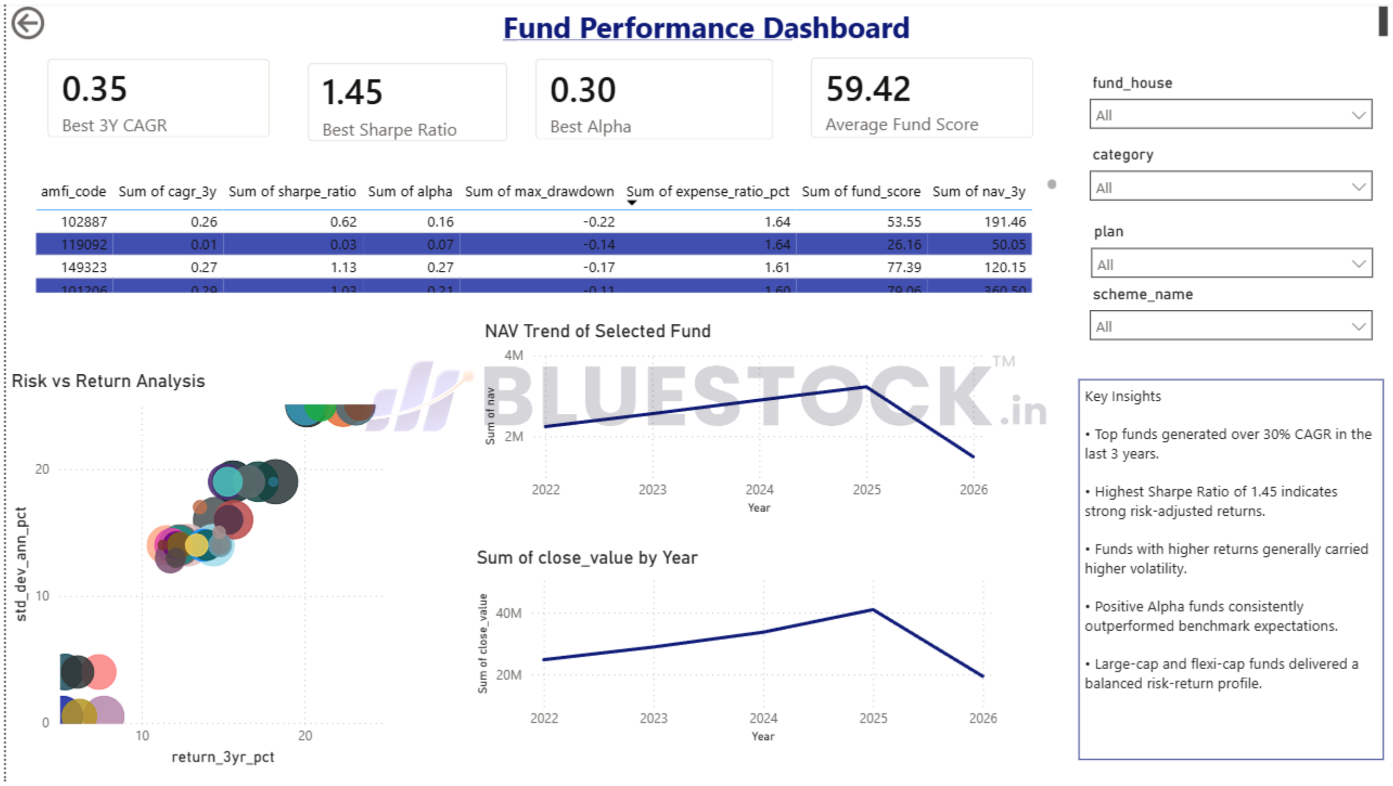
- Total AUM
- SIP Inflows
- Total Folios
- Number of Schemes
- Industry Trends
- Fund House Analysis



Page 2 – Fund Performance

Features:

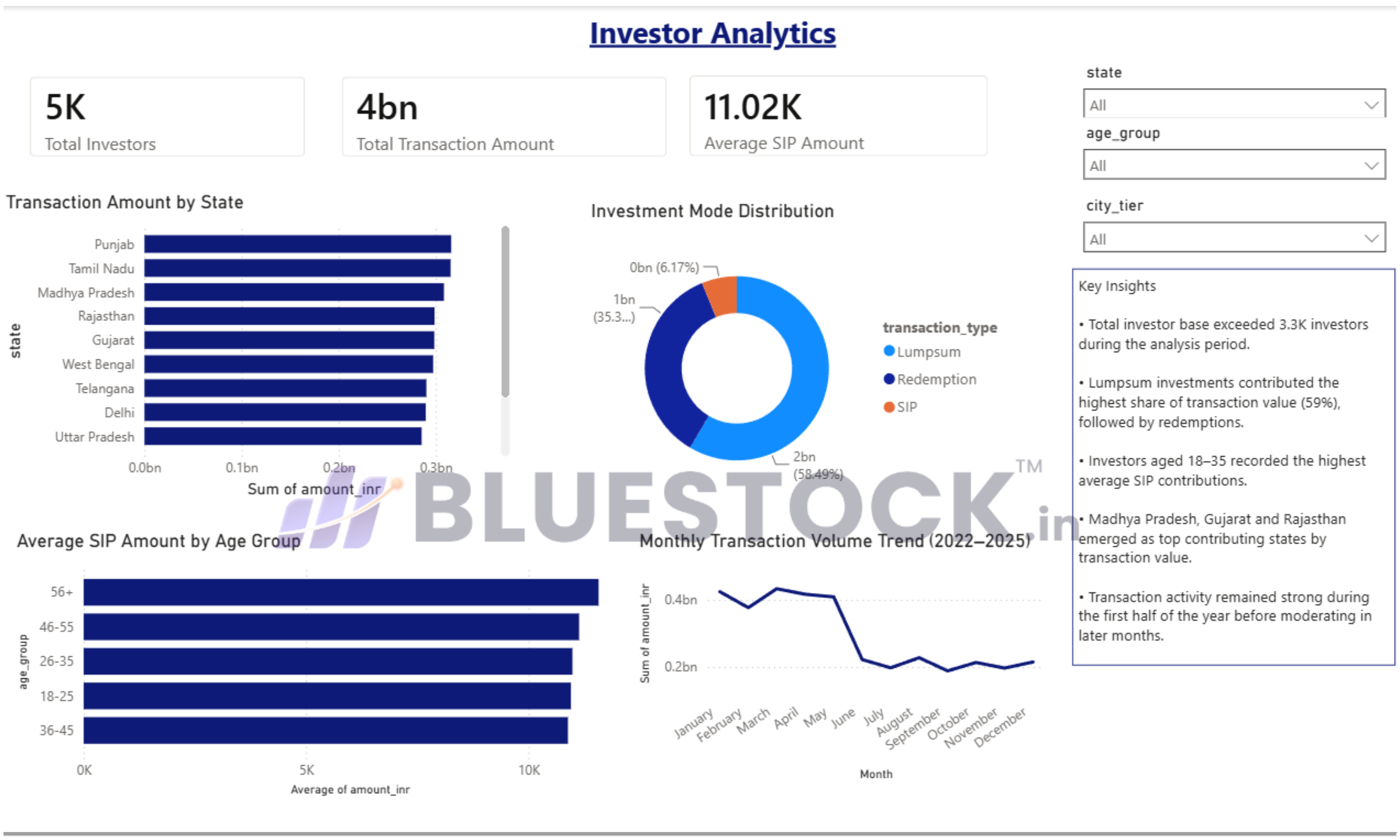
- Risk vs Return Analysis
- Fund Scorecard
- NAV Trends
- Benchmark Comparison
- Performance Filters



Page 3 – Investor Analytics

Features:

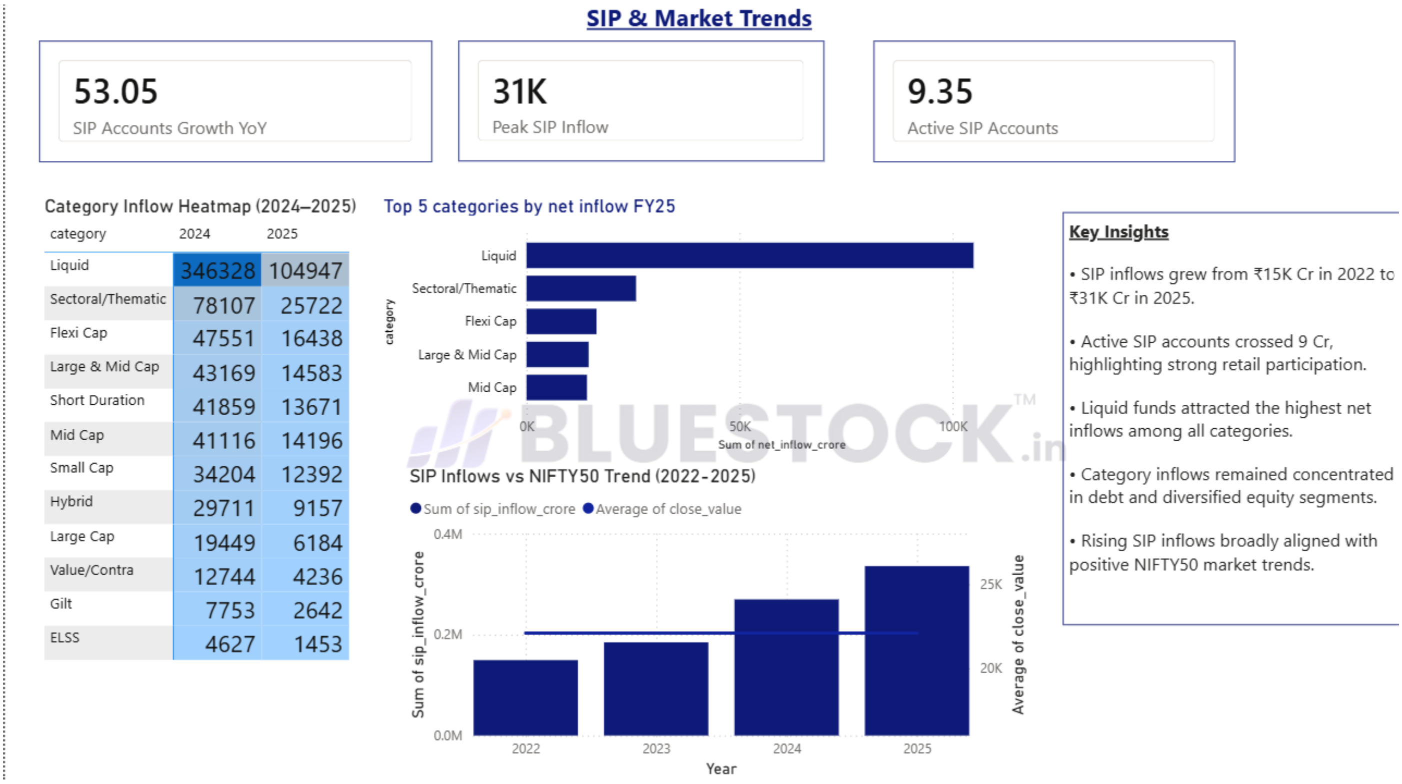
- State-wise Investments
- SIP vs Lumpsum Analysis
- Age Group Analysis
- Monthly Transaction Trends



Page 4 – SIP & Market Trends

Features:

- SIP Inflows vs NIFTY50
- Category Inflow Heatmap
- SIP Growth Analysis
- Market Trend Analysis



category	2024	2025
Liquid	346328	104947
Sectoral/Thematic	78107	25722
Flexi Cap	47551	16438
Large & Mid Cap	43169	14583
Short Duration	41859	13671
Mid Cap	41116	14196
Small Cap	34204	12392
Hybrid	29711	9157
Large Cap	19449	6184
Value/Contra	12744	4236
Gilt	7753	2642
ELSS	4627	1453

Key Findings

1. Mutual fund industry assets grew consistently throughout the analysis period.
2. SIP inflows reached record levels, indicating increasing retail participation.
3. Small-cap funds generated higher returns but also carried higher downside risk.
4. Rolling Sharpe analysis revealed significant changes in risk-adjusted performance over time.
5. Investor participation increased substantially in recent years.
6. SIP continuity analysis identified a large proportion of potentially inactive investors.
7. Diversification levels varied significantly across funds based on HHI analysis.
8. The recommendation engine successfully identified top-performing funds for different risk profiles.

Limitations

Several limitations should be considered:

- Dataset represents a limited sample of mutual fund schemes.
- Market conditions may influence performance metrics.
- Recommendation engine uses a simplified rule-based approach.
- Historical performance does not guarantee future returns.
- Synthetic data may not fully reflect real-world investor behavior.

Recommendations

1. Expand coverage to include additional mutual fund schemes.
2. Integrate live AMFI APIs for real-time updates.
3. Develop machine learning-based recommendation models.
4. Implement portfolio optimization techniques.
5. Add investor churn prediction models.
6. Deploy dashboard using cloud infrastructure.
7. Introduce automated reporting workflows.

Conclusion

This project successfully demonstrates the application of data analytics, financial analysis, and business intelligence techniques within the mutual fund domain.

The platform provides valuable insights into fund performance, investor behavior, risk management, and market trends. By combining Python, SQL, SQLite, and Power BI, the project delivers a complete analytical solution capable of supporting investment analysis and decision-making.

The project serves as a strong foundation for future enhancements involving predictive analytics, portfolio optimization, and advanced recommendation systems.